Compound 2D Shapes

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

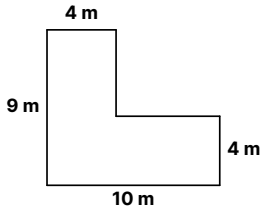
KEY WORDS composite shape · perimeter *(paenga)* · area *(horahanga)* · rectangle *(tapawhā hāngai)*

A Shapes made from rectangles ●○○

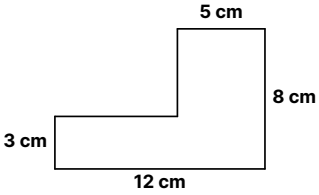
Remember: Find the missing side lengths first. For the area, split the shape into rectangles.

1. Find the perimeter and the area of each shape. All corners are right angles.

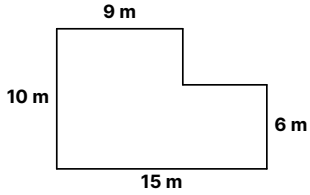
a)



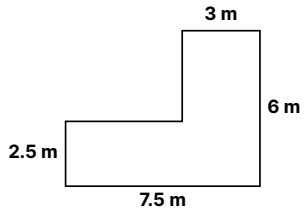
b)



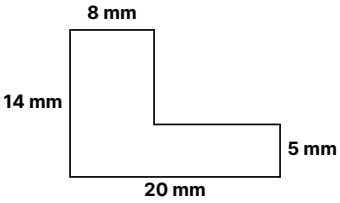
c)



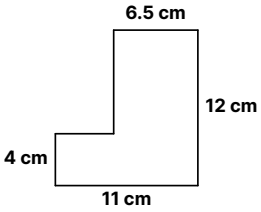
d)



e)



f)

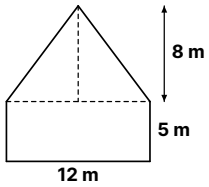


B Shapes with triangles ●●○

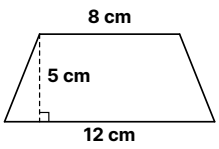
Remember: Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$. Area of a trapezium $= \frac{1}{2}(a + b) \times h$.

2. Find the area of each shape.

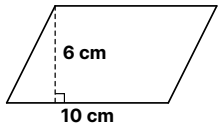
a)



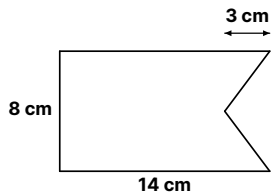
b)



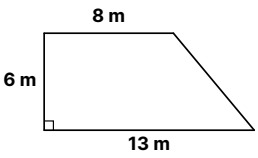
c)



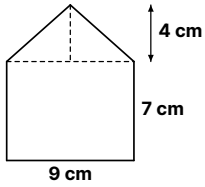
d)



e)



f)



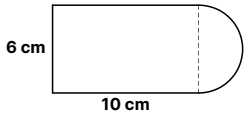
© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C Shapes with circles ●●●

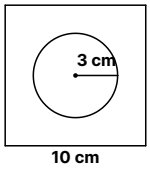
Remember: Area of a circle $= \pi r^2$. Circumference $= \pi d$. Give answers to 1 decimal place.

3. Find the area of each shape. In (b) the circle has been cut out. In (d) find the area inside the square but outside the circle.

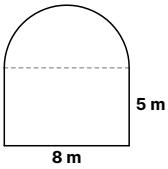
a)



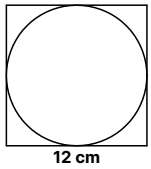
b)



c)



d)



4. Now find the perimeter of:

a) shape (a)

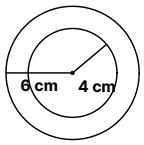
b) shape (c)

D Shaded regions ●●●

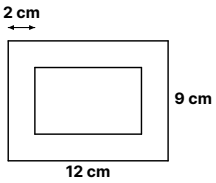
Remember: A shaded region is the big shape minus the hole. Find each area separately, then subtract.

5. Find each shaded area, to 1 decimal place where needed.

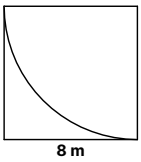
a)



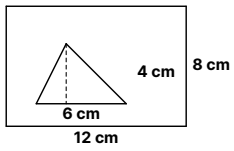
b)



c)



d)



E Problems ●●●

6. A rectangular lawn is 12 m long and 8 m wide. A circular flower bed with a radius of 2 m is cut out of it.

a) Find the area of the grass, to 1 decimal place.

b) Fertiliser for the grass costs $3.50 for each square metre. Find the cost of fertilising the lawn.

7. A wall is 6 m long and 2.5 m high. It has a window 1.2 m by 1.5 m and a door 0.8 m by 2 m. Find the area of wall to be painted.

8. A running track has two straight sides, each 100 m long, joined by a semicircle at each end. The semicircles have a diameter of 60 m. Find the distance once around the track, to the nearest metre.

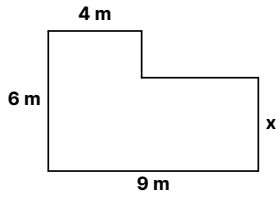
9. A round table top with a diameter of 1.2 m is cut from a square board with sides of 1.2 m. Find the area of board that is wasted, to 2 decimal places.

★ Extension

10. A square has sides of 10 cm. A quarter circle with a radius of 5 cm is cut from each of its four corners. Find the area that is left, to 1 decimal place.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

11. The L-shape has an area of 44 m². Find the length marked x.



© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

This page is blank so that the next version starts on a new sheet of paper.

Compound 2D Shapes

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

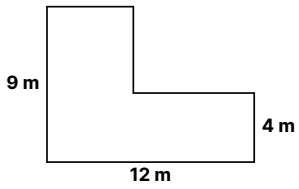
KEY WORDS composite shape · perimeter *(paenga)* · area *(horahanga)* · rectangle *(tapawhā hāngai)*

A Shapes made from rectangles ●○○

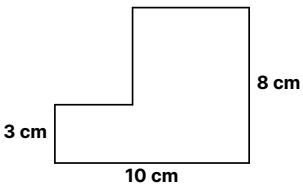
Remember: Find the missing side lengths first. For the area, split the shape into rectangles.

1. Find the perimeter and the area of each shape. All corners are right angles.

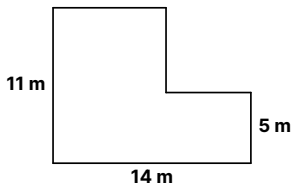
a)



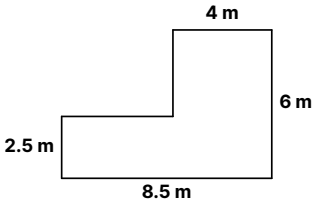
b)



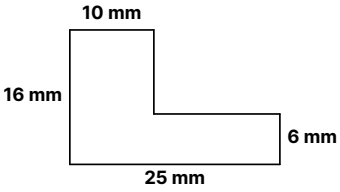
c)



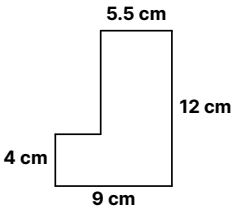
d)



e)



f)

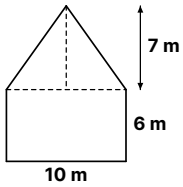


B Shapes with triangles ●●○

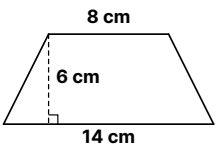
Remember: Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$. Area of a trapezium $= \frac{1}{2}(a+b) \times h$.

2. Find the area of each shape.

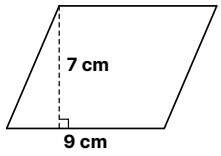
a)



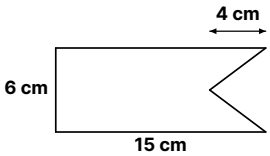
b)



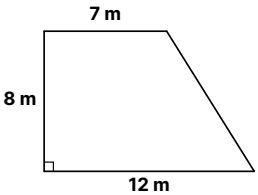
c)



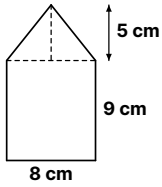
d)



e)



f)



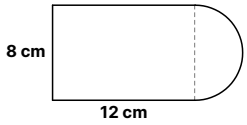
© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C Shapes with circles ●●●

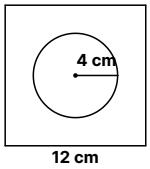
Remember: Area of a circle $= \pi r^2$. Circumference $= \pi d$. Give answers to 1 decimal place.

3. Find the area of each shape. In (b) the circle has been cut out. In (d) find the area inside the square but outside the circle.

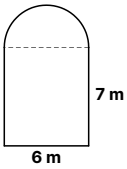
a)



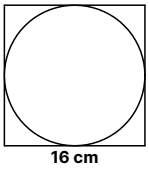
b)



c)



d)



4. Now find the perimeter of:

a) shape (a)

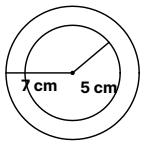
b) shape (c)

D Shaded regions ●●●

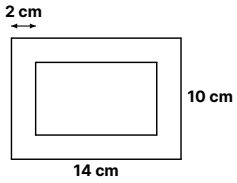
Remember: A shaded region is the big shape minus the hole. Find each area separately, then subtract.

5. Find each shaded area, to 1 decimal place where needed.

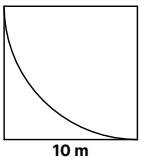
a)



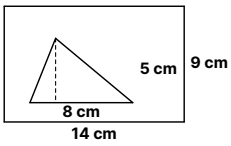
b)



c)



d)



E Problems ●●●

6. A rectangular lawn is 15 m long and 9 m wide. A circular flower bed with a radius of 3 m is cut out of it.

a) Find the area of the grass, to 1 decimal place.

b) Fertiliser for the grass costs $4.50 for each square metre. Find the cost of fertilising the lawn.

7. A wall is 7 m long and 2.4 m high. It has a window 1.5 m by 1.2 m and a door 0.9 m by 2 m. Find the area of wall to be painted.

8. A running track has two straight sides, each 80 m long, joined by a semicircle at each end. The semicircles have a diameter of 50 m. Find the distance once around the track, to the nearest metre.

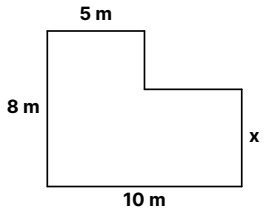
9. A round table top with a diameter of 1.4 m is cut from a square board with sides of 1.4 m. Find the area of board that is wasted, to 2 decimal places.

★ Extension

10. A square has sides of 12 cm. A quarter circle with a radius of 6 cm is cut from each of its four corners. Find the area that is left, to 1 decimal place.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

11. The L-shape has an area of 65 m². Find the length marked x.



© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.